Abhay Kabdwal

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Research Interests

PhD scholar at IIT Delhi working on privacy and anonymity systems, with a focus on mixnets and VPNs. My earlier work spans Artificial Intelligence, Quantum Machine Learning, Cryptography, and Digital Image Processing, and I am interested in how these areas can address data security challenges.

Education

Ph.D. Computer Science and Engineering
Indian Institute of Technology Delhi, New Delhi Jul 2026 - Present
Advanced Networks Lab, supervised by Prof. Piyush Kumar

M.Sc. Computer Science
University of Delhi, New Delhi 2024 - 2026

B.Sc. (Hons.) Computer Science
Banaras Hindu University, Varanasi Jan 2022 - Jun 2024
Cumulative Grade Point Average (CGPA): **9.12 / 10**

Senior Secondary (12th)
Jawahar Navodaya Vidyalaya, Nainital 2019 - 2021
Scored: **92% (CBSE)**

Current Research

- **Attacks on mixnets:** studying attacks on the Nym mixnet.
- **VPN auditing:** building an audit system for browser-extension based VPNs.

Scholastic Achievements

- **UGC NET (JRF):** Successfully qualified for the University Grants Commission National Eligibility Test 2025 for JRF and lectureship qualification with 99.89 percentile.
- **GATE Qualification:** Successfully qualified the Graduate Aptitude Test in Engineering (GATE) in 2025 and 2026 in Computer Science and Engineering (CSE).

Technical Competencies

- **Programming Languages:** Python, C++, JavaScript
- **Technologies & Tools:** Git, Linux, LaTeX, Overleaf
- **Web Development:** HTML, CSS, Flask, DBMS
- **Core Domains:** Privacy and Anonymity Systems, Deep Learning, Generative AI, Quantum Machine Learning, Cryptography, Image Processing

Past Projects

Project Reports Repository

Quantum Encoding for Image Compression (Completed) *2026*
Under the supervision of Dr. Om Pal, this research developed a novel hybrid method for Quantum Image Compression. Standard encoding frameworks like the Flexible Representation of Quantum Images (FRQI) and Novel Enhanced Quantum Representation (NEQR) face significant practical limitations due to massive qubit requirements. Our team collaboratively investigated optimization strategies that merge classical data compression with quantum state mapping to reduce these heavy demands.

Evaluating Hybrid Quantum-Classical CNNs (Completed) *fall - 2025*
Guided by Prof. Neelima Gupta and Dr. Om Pal, this research evaluated pure, hybrid, and multi-branch Quantum Convolutional Neural Network (QCNN) architectures across datasets like MNIST and COVID-19 radiography.
Project Report

Deep Learning Image Steganography (Completed) *spring - 2025*
Working alongside peers, this project explored deep learning methods to hide secret images within a standard cover image. We developed an encoder-decoder CNN architecture and trained it on the Tiny ImageNet dataset. **Project Report**

Hybrid Movie Recommendation System (Completed) *fall - 2024*
In this collaborative project, we explored the recommendation system domain and built a robust movie recommendation engine to overcome industry challenges like data sparsity and the "cold-start" problem. We initially implemented a Singular Value Decomposition (SVD) baseline, yielding a Normalized Discounted Cumulative Gain (NDCG) score of 0.81, but it struggled with complex user preferences. To improve this, we engineered a Neural Collaborative Filtering (NCF) framework combining Generalized Matrix Factorization with a Multi-Layer Perceptron (MLP). This successfully captured better user-item interactions, improving our NDCG score to 0.89. **Project Report**

Automated Legal Document Summarizer (Completed) *spring - 2024*
Under the guidance of Dr. Anshul Verma, this project involved a comprehensive evaluation of state-of-the-art abstractive summarization models applied to legal texts. Rather than developing a new architecture, the research explored and compared the performance of Legal Pegasus, BART, and the Longformer Encoder-Decoder (LED) on Indian and UK Supreme Court case datasets.

Certifications & Extracurriculars

- **Certifications:** Machine Learning Specialization by deeplearning.ai (Coursera - Andrew Ng).
- **Other Service:** Website Committee Member (2024 - Present) for the Department of Computer Science, University of Delhi, actively maintaining and updating the official website.
- **Teaching / Content Creation:** YouTube Content Creator (2022 - 2024), producing and publishing video lectures to simplify core computer science concepts for a broad online audience. **YouTube Channel**
- **Sports & Athletics:** Four-time participant in Navodaya Vidyalaya Samiti (NVS) National Level Games and Two-time participant in School Gaming Federation of India (SGFI) representing NVS as a state for the sport Yoga (2016 - 2019), showing athletic dedication.
- **Interests:** Teaching, Open-source contributions, Tech Blogging, Calisthenics, Gym, Chess.